

Root Cause Graph — Analytical Methodology

Causal fault-propagation, alarm correlation, and root-cause ranking · NEXUS-1 analytics layer

Purpose of this note

The Root Cause Graph is the most method-bearing screen in the platform: unlike the read-once monitoring views, it makes explicit claims about how a cause is inferred. This note records the reasoning behind those claims — what the graph is, why it is built the way it is, and, equally important, what it deliberately does not do. It is a rationale document, not a UI walk-through; the in-application Help entry covers navigation.

The three-layer model

Causal structure in the graph is **engineered, not discovered from telemetry**. This is the central design decision. Inferring causal edges from observational time-series is unreliable under exactly the conditions a reactor presents — tight control-loop feedback, confounders, and sensor lag routinely manufacture spurious edges. The graph therefore rests on an auditable backbone, with data used only to weight and propose, never to define.

Layer	Source	Role
1 · Engineered backbone	Plant fault trees / event trees (PRA-PSA) and component topology	Defines the causal edges. Every edge is traceable to an analysed failure path — auditable and regulator-aligned.
2 · Live runtime overlay	Active alarms and anomaly flags	Lights nodes by severity and animates propagation along the known edges. Adds state, not structure.
3 · Data-informed layer	Learned weights / candidate edges (ML)	Adjusts edge confidence and proposes candidate edges for engineer confirmation. Hints only — held out of the backbone until confirmed.

On screen these read as: **solid** edges = fault-tree backbone; **dashed violet** = data-informed weight; **struck grey** = a candidate edge proposed from correlation and then rejected. Node borders encode severity, not cause: **red** = origin or critical cascade, **amber** = degraded / transmitting, **green** = healthy or ruled out.

Worked example — incident EVT-2026-0418

A feedwater valve develops actuator latency. The graph reconstructs the propagation from that origin to the protective trip four hops downstream:

Valve FV-104 (actuator latency) → SG-1 Level (low → low-low) → FW Pump 2A (overspeed +12%) → Pump 2A Bearing (temp + vib) → Primary Loop Flow (0.8 Hz oscillation) → Core delta-T (>3σ → >50°C) → REACTOR TRIP (17:12:14)

Two paths converge on the loop-flow node: the feedwater degradation above, and an independent **RCP-1B** vibration entering in parallel — which is why loop flow is a genuine convergence point rather than a single-cause symptom. A **data-informed** edge from **4kV Bus 2A** (voltage transient) re-weights pump stress as a low-confidence contributor, and a **rejected** candidate edge from **Flow Sensor FT-7** records the hypothesis “is this just a sensor fault?” being considered and discarded — the sensor cross-checks healthy. The terminal effect is the reactor trip; its immediate trigger is the high core delta-T, four hops from the origin. The trip is the protection system working correctly — the analytical value is in walking back up the chain to the valve.

Payoff 1 — alarm-flood suppression

The incident raised **14 alarms in just over four minutes**. The originating fault (FV-104) is a single low-severity alarm buried mid-flood, while the loud downstream alarms — bearing, loop flow, core delta-T — dominate attention. The propagation graph collapses the flood into one origin plus its grouped downstream cascade, suppressing roughly 93% of the triage noise. This is the ISA-18.2 alarm-management problem, and the difference between an operator drowning and an operator oriented.

Payoff 2 — causal-origin ranking beats alarm-count ranking

The strongest demonstration is a reversal. Ranking hypotheses by **alarm count** puts the most-alarmed node — the pump bearing — on top; acting on it treats a symptom while the upstream valve fault recurs. Ranking by **causal origin** traces the graph upstream and promotes FV-104, which fired only one alarm yet explains twelve of fourteen.

Hypothesis	Role	Direct alarms	Explains	Edge	Causal confidence
FV-104 actuator latency	ORIGIN	1	12 / 14	fault-tree	66%
Pump 2A bearing degradation	PROXIMATE	2	4 / 14	fault-tree	20%
RCP-1B bearing wear	PARALLEL	2	3 / 14	fault-tree	7%
4kV bus voltage transient	CONTRIBUTING	1	3 / 14	data-informed	5%
Flow sensor FT-7 fault	RULED OUT	0	0 / 14	rejected	2%

Confidences are illustrative, derived from edge weights, and are not identified from plant data.

Limitations & non-claims

- The causal structure is engineered from fault/event trees — the platform does **not** perform causal discovery from telemetry, and learned edges never enter the backbone unconfirmed.
- Confidence scores are illustrative weightings for ranking and triage, not calibrated probabilities or a PRA result.
- The graph is an **advisory and investigative** aid. It informs operators and investigators only; it has no control authority and never interacts with certified, hardware-isolated reactor protection systems.
- Output is a hypothesis set for engineer confirmation. A confirmed root cause feeds the incident record and the hash-sealed compliance log; the graph does not close an investigation on its own.